Whole Body Control of a Wheeled Quadruped using Model Predictive Path Integral Controll

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***Abstract*—We present a real–time whole-body motion–planning framework that extends Model Predictive Path Integral (MPPI) control from purely legged quadrupeds to *wheeled–legged* robots. Building on the Unitree Go2W platform, we augment the action space with four wheel–torque channels and enrich the running cost with (i) wheel-velocity regulation, (ii) a PD-shaped joint effort term, and (iii) an L_1 penalty on base positional drift. A wheel-aware modification of the Raibert heuristic guides joint–angle sampling, while soft joint-limit penalties maintain mechanical feasibility.**

Simulation studies demonstrate agile behaviors that are difficult for purely legged systems: rolling stair ascent, fast rough-terrain traversal, a stable $35\ \mathrm{cm}$ box jump, and waypoint following on flat ground. Compared with a leg-only baseline, the wheeled controller reduces traversal time by 55.7% and achieves 134.5% greater forward velocity on rough terrain while keeping wheel–ground contact smooth. The entire pipeline operates without offline learning or precomputed contact schedules, highlighting MPPI's ability to handle the hybrid dynamics of wheeled–legged locomotion in a uniform sampling-based fashion.

These results indicate that sampling-based MPC can serve as a unified control layer for future multi-modal robots that must seamlessly switch between walking, rolling, and jumping in unstructured environments. Link to the source code can be found here: (https://github.com/kenryhou2/MPPI_quad)

I. Introduction

EXPLORATION of the world around us is a critical motivator of modern field robotics research. Unlike the controlled lab environments used to test well-structured functionality of robotic platforms, environmental exploration is much more unstructured and presents an exciting challenge for robot performance. In particular, robotic platforms must contend with uncertainties in contact, obstacle traversal, and complex terrain.

A promising platform for unstructured and rough terrain navigation is the wheeled quadruped. These systems have the potential to operate at much higher speeds than their legged counterparts in complex terrain [1], and can more readily contend with obstacles with their multi-modal locomotion capabilities. Recently, there has been interest in motion planning [2] and rough terrain traversal [3], [4] for wheeled quadrupedal robots. However, these studies focused on augmenting the rolling aspect of these platforms while relying on the legs as an active suspension to traverse uneven terrain. These studies do not plan across tasks that require whole-body movement, such as box jumping and stair climbing.

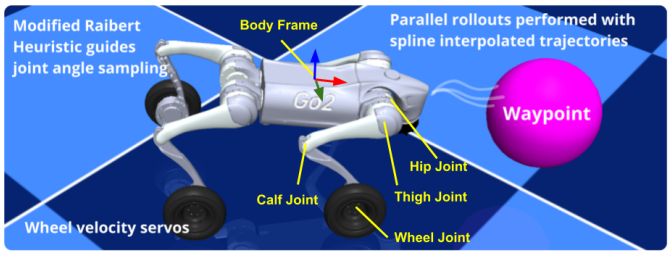


Fig. 1: Overview of the wheeled-quadruped whole body control pipeline. MPPI performs parallel rollouts using spline-interpolated joint trajectories guided by a modified Raibert heuristic. Joint-level torque control is applied to the hip, thigh, and calf joints, while wheel motion is governed by velocity servos. The body frame origin is fixed on the center body and serves as the reference for motion planning.Videos of locomotion can be viewed at the following link: (https://drive.google.com/drive/folders/1-P6Oea_fGQhj3ui2CDeBRzEe1X1tqOHc?usp=sharing)

Simultaneously, however, there have been exciting new developments with whole-body locomotion and manipulation using MPPI and its deployment on real-world platforms [5]. These developments showcase the potential of whole-body control schemes in executing traditionally difficult locomotion tasks, such as stair climbing and box jumping. In this paper, we seek to expand the implementation of full body MPPI by Alvarez et al. to include velocity controlled wheels at the ends of each leg, and demonstrate its application to the previously mentioned locomotion tasks. We model this robot in simulation with a Unitree Go2W robot. The key changes that we have made to the whole-body planner to account for the added wheels is to include a L_1 penalty on base positional drift, a PD-shaped joint effort term, a modification to the Raibert heuristic, and a regulator tracking wheel velocities. With these modifications, we can demonstrate in simulation agile behaviors, such as stair climbing and rough terrain traversal, that exceed the performance of legged quadrupeds in the same scenario.

The remainder of this paper is organized as follows: We first review the fundamentals of MPPI and trajectory sampling in II-A and II-B. We then delve into our methods and modifications to the original whole-body MPPI in III. We then discuss the results of our task simulations in IV and discuss limitations and future work in V.

II. Background and Related Work

A. Model Predictive Path Integral (MPPI) Controller

The high-level locomotion policy is implemented as a Model Predictive Path Integral (MPPI) controller [6], executed on top of the MuJoCo simulator described in Section III-A. At every control cycle the algorithm draws N action sequences $\{\mathbf{U}^{(i)} \in \mathbb{R}^{T\times m}\}_{i=1}^{N}$, rolls out the system for a horizon of T steps with time step h, and updates the nominal control trajectory via importance weighting. The present implementation extends the legged formulation of Alvarez-Padilla *et al.* [**?**] to a wheeled–legged quadruped by (i) appending four wheel-torque channels to the action vector and (ii) augmenting the running cost with explicit wheel-velocity regulation. A schematic overview is given in Alg. 1.

Algorithm 1 MPPI Control Algorithm

Input: current state $\mathbf{x}_t$, nominal control trajectory $\bar{\mathbf{U}}$
for $i = 1, \ldots, N$ **do**
  sample $\Delta\mathbf{U}^{(i)} \sim \mathcal{N}(0, \Sigma)$ (Section II-B)
  $\mathbf{U}^{(i)} \leftarrow \mathrm{clip}(\bar{\mathbf{U}} + \Delta\mathbf{U}^{(i)},\ \mathbf{u}_{\min}, \mathbf{u}_{\max})$
  rollout $\mathbf{x}_{t:t+T}^{(i)}$ with MuJoCo
  compute cost $J^{(i)}$ using (4)
end for
$\omega^{(i)} = \exp\!\big(-\frac{1}{\lambda}(J^{(i)} - \min_j J^{(j)})\big)$
$\bar{\mathbf{U}} \leftarrow \sum_{i=1}^{N} \omega^{(i)}\mathbf{U}^{(i)} \big/ \sum_{i=1}^{N} \omega^{(i)}$
Output: first control element $\bar{\mathbf{u}}_0$

B. Trajectory Parameterization and Sampling

Action sequences are generated by perturbing the nominal trajectory with zero-mean Gaussian noise. Two sampling schemes are supported:

1) *Normal sampling*: independent noise is drawn for every time step, yielding an $N \times T \times m$ tensor of disturbances.
2) *Cubic-spline sampling*: noise is sampled at K knot points ($K \ll T$) and interpolated with a cubic spline to produce temporally correlated perturbations. This reduces the effective search dimensionality and improves sample efficiency for large horizons (K=8 and T=40 in the reported experiments).

The noise covariance $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_m^2)$ is time-invariant and adjusted online according to the commanded gait (Section III-E). Joint-space limits $\mathbf{u}_{\min}, \mathbf{u}_{\max}$ are enforced by clipping the perturbed trajectories before simulation, guaranteeing dynamically feasible rollouts.

C. Go2-W Wheeled Quadruped

The simulated platform is based on the Unitree Go2W wheeled quadruped, modeled in MuJoCo with a floating base frame. Each leg is composed of a hip (abduction), thigh (hip pitch), and calf (knee) joint, culminating in a torque-actuated wheel also depicted in Fig. 1. These joints are modeled explicitly and controlled via MuJoCo's torque actuators. The wheels are mounted coaxially to the calf links and actuated along the lateral axis, enabling smooth rolling locomotion. All links and inertias are defined in the URDF/MJCF model, and contact dynamics are resolved using MuJoCo's native collision model and linearized friction cones.

D. Related Work

Recent work in wheeled–legged locomotion has advanced both model-based and learning-based control strategies, each with distinct trade-offs compared to MPPI. Bjelonic et al. [2] proposed a whole-body motion planner for torque-controlled wheeled quadrupeds using a ZMP-based trajectory optimizer coupled with a hierarchical whole-body controller based on solving prioritized quadratic programming problems. While their framework handles nonholonomic constraints and enables hybrid locomotion over uneven terrain, it relies on a simplified model and suffers from limited adaptability during execution due to reliance on precomputed plans. Medeiros et al. [7] developed a trajectory optimization pipeline incorporating terrain geometry and contact forces in an NLP framework. Their method enables dynamic driving on rough terrain but is computationally expensive and not suited for real-time receding-horizon execution. In contrast, Lee et al. [8] used model-free reinforcement learning to train a hybrid locomotion controller integrated with a learned navigation policy. Though this achieves robust autonomy in large-scale urban settings, it requires extensive simulation training and lacks interpretability or guarantees on constraint satisfaction. Compared to these approaches, our MPPI framework offers a real-time, sampling-based alternative that balances control fidelity and responsiveness without requiring pre-training.

III. Methods

A. MuJoCo Physics Engine for MPPI Control

The approach of our whole-body sampling controller is based on a simulator that can *both* (i) execute many trajectories in parallel and (ii) advance each individual simulation step fast enough to close a real–time feedback loop. We therefore adopt the CPU implementation of MuJoCo [9], which strikes a balance: on a recent high-end processor (e.g. AMD Ryzen Threadripper Pro) we obtain 30–50 parallel dynamics rollouts whose wall-clock step time comfortably supports a 0.01 s discretization and a 0.4-0.8 s prediction horizon. This level of parallelism is sufficient for the tasks considered in this work, yet avoids the end-to-end latency penalties that arise when waiting for thousands of GPU rollouts to complete. Each rollout uses the default contact parameters in MuJoCo and a linearized friction cone approximation to accelerate contact resolution. All rollouts are spawned in separate threads and

synchronized at the end of every MPC iteration to compute trajectory costs and importance weights. With these settings, the overall controller—including spline sampling, simulation, cost evaluation, and weight update—runs at ≈ 100 Hz, enabling real-time whole-body optimization directly on hardware without any offline training or contact pre-specification.

B. Augmented Running Cost

For each sampled trajectory $i \in \{1, \ldots, N\}$ and horizon index $k = 0, \ldots, T-1$ we form three error terms.

(1) State-tracking error: Only orientation, leg-joint angles, and wheel velocities are tracked. Cartesian position is handled separately later.

$$\Delta \mathbf{x}_k^{(i)} = \begin{bmatrix} \mathbf{0}_3 \\ d_q\big(\mathbf{q}_k^{(i)}, \mathbf{q}_{k,\mathrm{ref}}\big)\,\mathbf{1}_4 \\ \mathbf{q}_k^{\mathrm{leg}} - \mathbf{q}_{k,\mathrm{ref}}^{\mathrm{leg}} \\ \boldsymbol{\omega}_k^{\mathrm{wheel}} - \boldsymbol{\omega}_{k,\mathrm{ref}}^{\mathrm{wheel}} \end{bmatrix}. \tag{1}$$

(2) PD-shaped control error: Only the first twelve torque channels (leg actuators) are penalized:

$$\Delta \mathbf{u}_k^{(i)} = K_p\big(\boldsymbol{\tau}_k^{\mathrm{leg}} - \mathbf{q}_k^{\mathrm{leg}}\big) - K_d\,\dot{\mathbf{q}}_k^{\mathrm{leg}}, \quad K_p = 82,\ K_d = 12. \tag{2}$$

Wheel-torque channels remain unpenalized; their regulation is enforced implicitly through wheel-velocity tracking in (1).

(3) Cartesian position error (L_1 norm):

$$\Delta \mathbf{p}_k^{(i)} = \mathbf{p}_k^{(i)} - \mathbf{p}_{k,\mathrm{ref}}, \qquad L_{1,k}^{(i)} = \big\|\Delta \mathbf{p}_k^{(i)}\big\|_1. \tag{3}$$

Trajectory cost: With positive-definite weight matrices $\mathbf{Q} \succ 0$ and $\mathbf{R} \succ 0$ the cost of roll-out i is

$$\begin{aligned} \mathcal{L}^{(i)} = &\sum_{k=0}^{T-1} (\Delta \mathbf{x}_k^{(i)})^\top \mathbf{Q}\, \Delta \mathbf{x}_k^{(i)} \\ &+ \sum_{k=0}^{T-1} (\Delta \mathbf{u}_k^{(i)})^\top \mathbf{R}\, \Delta \mathbf{u}_k^{(i)} \\ &+ \sum_{k=0}^{T-1} L_{1,k}^{(i)}. \end{aligned} \tag{4}$$

Equation (4) feeds directly into the MPPI importance weights (Alg. 1).

C. Joint-Limit Handling

Besides the pre-simulation clipping of actions, soft penalties are applied during cost evaluation to discourage prolonged operation near mechanical limits:

$$\ell_{\mathrm{lim}} = \alpha\,\big[\max(0, \mathbf{q} - \mathbf{q}_{\max})^2 + \max(0, \mathbf{q}_{\min} - \mathbf{q})^2\big], \tag{5}$$

with $\alpha = 10^3$ and bounds taken from the manufacturer's specifications. This term only activates when a limit is violated, leaving the interior of the admissible set unaffected.

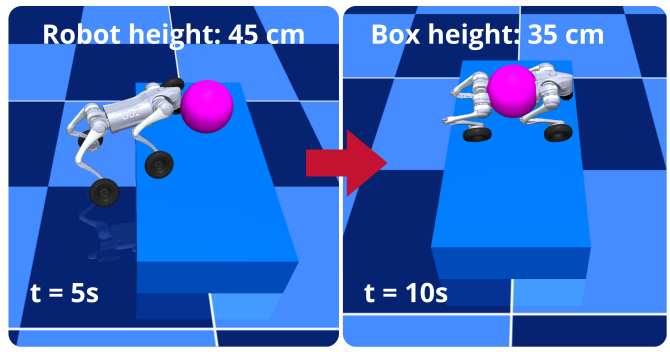


Fig. 2: Box-pushing scenario showing the robot dynamically interacting with a 35 cm obstacle. At $t = 5$ s, the robot approaches the object at a height of 45 cm; by $t = 10$ s, it successfully mounts the box while maintaining stability and waypoint tracking. The behavior emerges from the MPPI controller without explicit planning of footholds or terrain contact transitions.

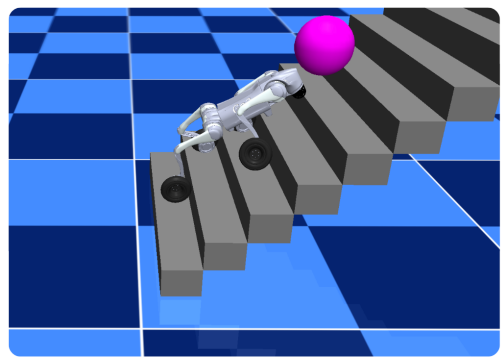


Fig. 3: Wheeled-quadruped robot climbing a 10-step staircase using MPPI control. Each step is 15 cm high and 20 cm deep.

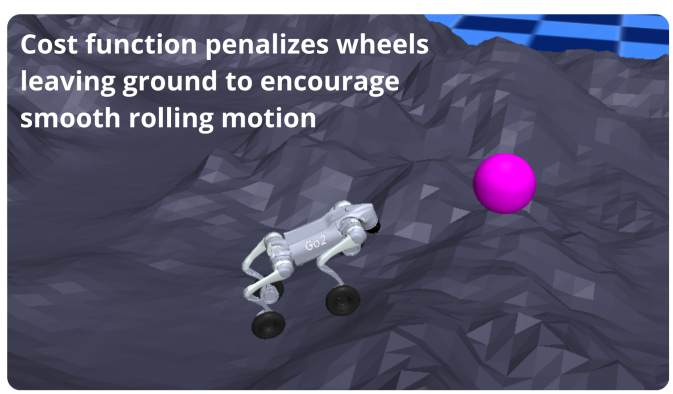
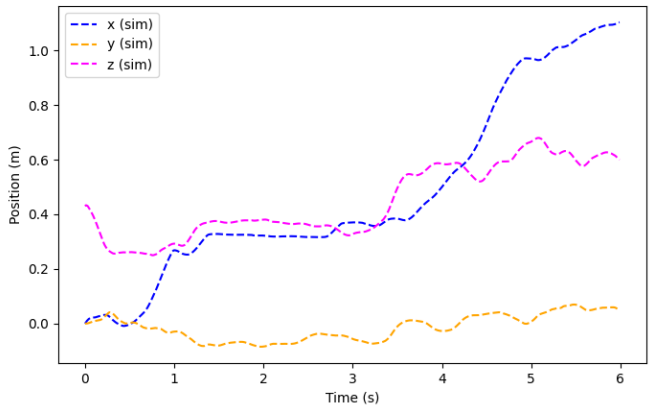
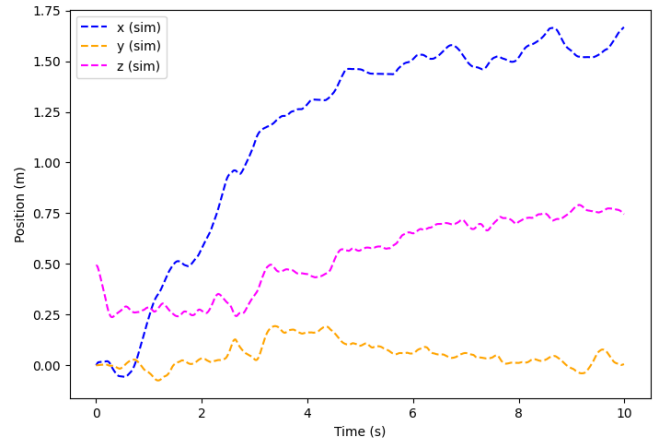


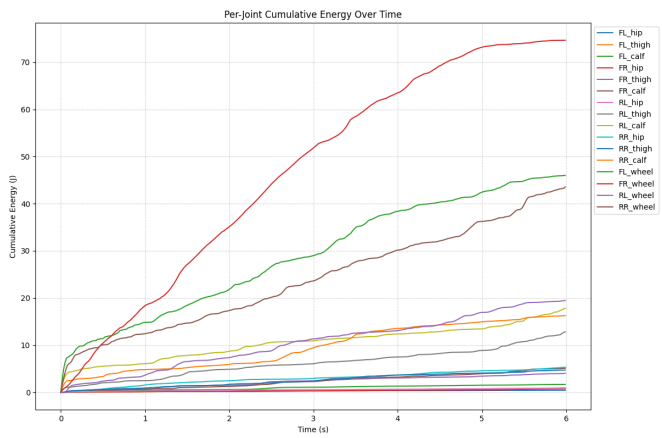
Fig. 4: Locomotion over rugged terrain with significant height variation of maximum 1m. The MPPI cost function includes a penalty for wheel-ground detachment, promoting smooth, ground-hugging trajectories. This regularization improves stability and contact persistence even under highly discontinuous terrain profiles.



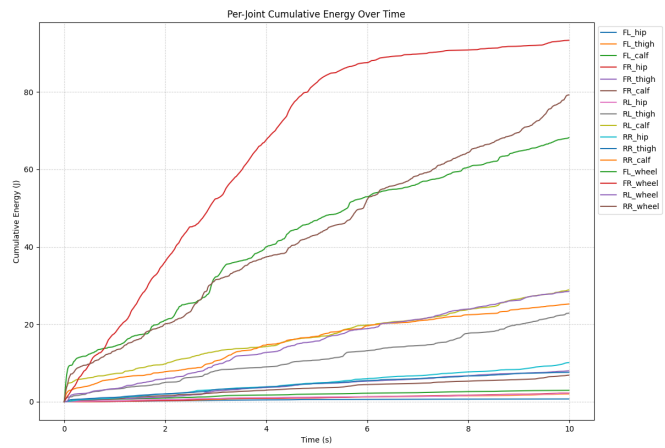
(a) Body Frame Position for Wheeled Quadruped - 35cm Box Jump



(b) Body Frame Position for Wheeled Quadruped - Stairs



(c) Joint Level Total Energy for Wheeled Quadruped - 35cm Box Jump



(d) Joint Level Total Energy for Wheeled Quadruped - Stairs

Fig. 5: Body Frame Position and Joint Energy Profile Results for Box Jump and Stair Climb

D. Modified Raibert Heuristic

To anticipate the coupled wheel–leg dynamics during gait transitions, the touchdown locations prescribed by the original Raibert heuristic are modified by an offset proportional to the commanded wheel speed:

$$\Delta \mathbf{x}_{\text{foot}} = k_s \, \boldsymbol{\omega}^{\text{wheel}} \, h, \tag{6}$$

where k_s is an empirically tuned constant. The adjustment prevents premature wheel–ground contact when accelerating and improves net traction on slippery terrain. The offset is recomputed at each planning step, enabling quick adaptation to changing velocity commands.

E. Gait Scheduler and Internal References

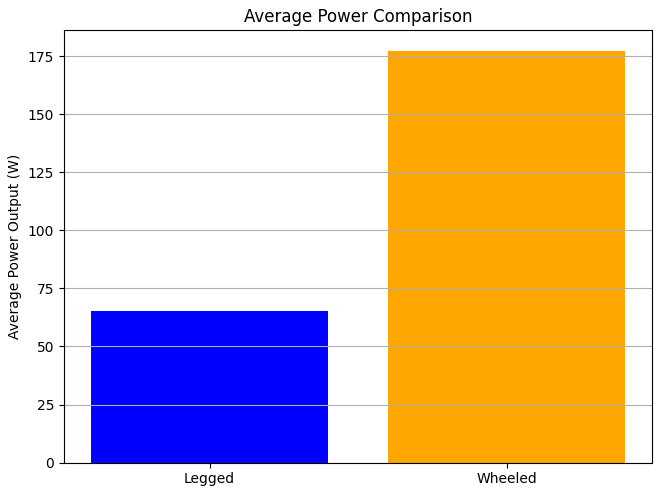
A finite-state scheduler selects among five template gaits—*in-place*, *walk*, *walk-fast*, *trot*, and *wheeled*—based on a predefined goal sequence. Each gait provides a $T \times 12$ matrix of desired leg-joint trajectories $\mathbf{q}_{\text{ref}}^{\text{leg}}(t)$ and a matched wheel The scheduler advances once a goal tolerance $\|\mathbf{p} - \mathbf{p}_{\text{goal}}\| < \varepsilon$ is met for a dwell time τ_{wait}. When the gait changes, the sampling covariance Σ is resized to reflect the differing control bandwidths of slow versus dynamic motions.

IV. Results

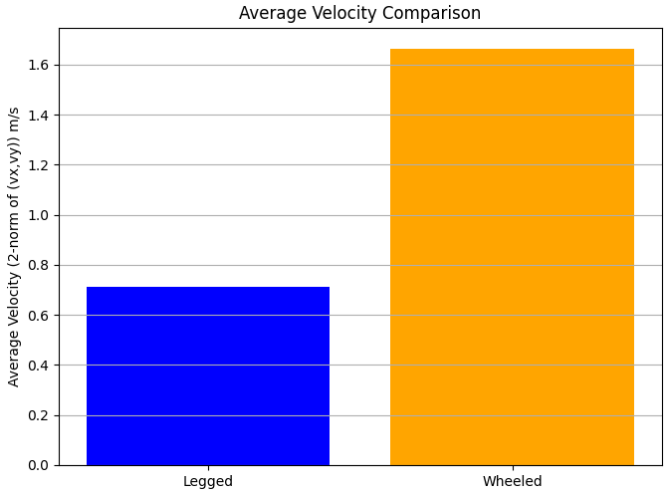
We evaluate the performance of our MPPI-controlled wheeled quadruped across three challenging scenarios—box jump, stair climbing, and rough terrain traversal—designed to assess dynamic locomotion, elevation handling, and contact robustness under discontinuous and irregular terrain conditions. Flat ground rolling walking gait is featured in the supplementary video found at (https://drive.google.com/drive/folders/1-P6Oea_fGQhj3ui2CDeBRzEe1X1tqOHc?usp=sharing)

A. Locomotion - Box Jump

The wheeled quadruped successfully completed a box jump maneuver (Fig. 2 within 6 s, reaching a platform height of 35 cm. Fig. 5a shows the base position trajectory over time, with the x and z components exhibiting smooth, stair-like elevation gains, indicating coordinated propulsion and lift



(a) Comparison of Average Power Output - Rough Terrain Traversal



(b) Comparison of Average Power Output - Rough Terrain Traversal

Fig. 6: Torque and Velocity Comparisons between Wheeled and Legged Quadruped - Rough Terrain Traversal

phases. Notably, the robot achieved a peak elevation exceeding 60 cm, providing sufficient clearance over the obstacle.

Velocity profiles reveal non-zero vertical acceleration spikes during takeoff and landing phases, consistent with short-duration flight dynamics. Although lateral oscillations are present in the y axis, the robot maintained directional stability throughout the maneuver.

Total energy expenditure over the 6 s trial (Fig. 5c) was approximately 255 J, with dominant energy contributions from the rear wheels (Fig. 5c), suggesting that propulsion was primarily wheel-driven. This aligns with the control strategy's emphasis on minimizing leg effort when possible.

The MPPI per-step cost displayed characteristic spikes during initial motion onset, takeoff, and landing, but converged to a stable low-cost regime once the robot stabilized atop the box. These results indicate that the controller effectively navigated discontinuous terrain while maintaining energy efficiency and trajectory tracking performance.

B. Locomotion - Stair Climbing

The wheeled quadruped demonstrated robust stair-climbing behavior over a 10 s staircase window, with each step measuring 15 cm in height and 20 cm in depth. Figure 5b shows the evolution of the base position over time. The z-axis trajectory exhibits clear step-wise elevation gains, consistent with successful traversal of each step, reaching a final height of approximately 75 cm. The x progression is steady, indicating consistent forward motion, while lateral deviation in y remains minimal.

The MPPI per-step running cost decreases sharply after initial contact transitions, then stabilizes in the midrange of 10^5–10^6 as the robot ascends. A brief spike around 2 seconds corresponds to the initiation of the first significant vertical displacement.

Energy analysis (Fig. 5d) indicates dominant consumption by the front and rear wheels, with peak values exceeding 85 J per joint. Notably, rear leg calf joints also contribute substantial energy, highlighting the combined use of rolling and lifting actions. Compared to the box-jump task, stair climbing exhibits a more distributed energy profile across joints, with increased involvement from the legs to overcome discrete elevation transitions. Together, these results confirm that MPPI control enables stable and energy-aware locomotion over discontinuous terrain, with adaptive modulation of joint effort during climbing phases.

C. Locomotion over Challenging Terrains

We compare the performance of legged and wheeled quadrupeds on a rough terrain map reminiscent of a Martian or rocky environment to investigate the advantage of wheeled locomotion. Both robots run the MPPI controller with the same hyperparameters (horizon, samples, rollouts, temperature). The terrain was simulated in the MuJoCo environment using a variable height map defining a single rigid body. The locomotion task was defined as following nine waypoints along the terrain, which were chosen manually to represent the reasoning of a traversability-aware planner that is often used in a full autonomy stack. Despite the waypoints being chosen to provide reasonably feasible paths for both robots, the wheeled quadruped achieves much higher performance, completing the task 55.7 % faster at (4.93 seconds vs. 11.12 seconds), but with 172.5 % higher average power output (177.5 watts vs 67.13 watts). It also achieves 134.5 % higher average translational velocity of 1.67 m/s (vs. 0.71 m/s). These results show the advantage of wheeled locomotion as a more adaptable alternative to traditional legged locomotion in highly variable and rugged environments, but currently requires more energy output.

V. Conclusion and Future Work

In this paper we presented an implementation of a whole-body MPPI controller on a wheeled quadruped in simulation. We demonstrated successful completions of tasks, such as stair climbing, box jumping, and rough terrain traversal, that replicated and in some cases exceeded the performance of conventional non-wheeled quadrupeds.

A. Limitations

Contact remains a prevalent problem with quadrupedal locomotion through rough terrain and is enhanced with the introduction of wheels. Unexpected contact with surfaces during walking gaits results in poor performance and behavior, which was particularly noticeable during stair climbing simulations. Additionally, our controller cannot be directly implemented on a real-time deployment in unstructured environments. Further research must be conducted on combining controllers with real-time environmental observation and modeling. Additionally, results from rough terrain traversal show significantly higher energy usage of the wheeled quadruped.

B. Future Work

There are many directions that warrant further exploration: First, we would like to explore structured deployment of our controller on physical platforms to verify the results of our simulation. Second, we would like to explore locomotive behaviors such as ”skating” in replacement of traditional behaviors like ”trotting”. Additionally, we would like to explore combining real-time whole-body control with real-time environmental observation and modeling, to allow the controller to build a map of its surrounding and optimize its trajectory across uncertain, unstructured environments. Finally, we would attempt to imporove energy efficiency of the wheeled locomotion by cost tuning exploring different rolling gaits.

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